Elementary Matrix Method for Dispersion Analysis in Optical Systems

Douglas M. Baney, *Senior Member, IEEE*, and Bogdan Szafraniec, *Member, IEEE*

(Invited Paper)

***Abstract*—In this paper, dispersion analysis of optical components and systems is presented using a formalism based on the elementary matrices and the N-matrix, first described by Jones. This approach readily incorporates both phase and amplitude dispersion in a generalized dispersion framework. The method simplifies the analysis of the combined effects of group delay, differential group delay, amplitude slope, and differential amplitude slope as compared to traditional Jones matrix methods. Higher order polarization-mode dispersion and the effects of concatenation are presented along with a discussion of measurement principles. The application of the elementary matrix concept to Mueller matrix methods in Stokes space is also discussed.**

***Index Terms*—Chromatic dispersion, differential amplitude slope, optical fiber measurements, optical network analysis, polarization-mode dispersion.**

I. Introduction

THE dependence of the propagation of light on the material and the structural properties of the medium it traverses is of great consequence in optical communications systems. The modulated optical waves used in fiber optic communications extend over many gigahertz of bandwidth, and the signals become impaired if the constituent spectral components exhibit different delays or amplitude effects. In addition to these chromatic dispersive effects, there is often a dependency on the polarization vector of light. The orientation of this vector may cause the original wave to split into multiple waves in the course of interaction with optical birefringence present in the system, resulting in impaired communications performance [1]–[10]. This effect is often described in terms of polarization-mode dispersion (PMD) and higher order PMD when there is a dependency of birefringence on optical frequency. Given the significant impact of dispersion on optical systems, a large amount of literature is available that provides analysis and measurement of various aspects of dispersion [11]–[13].

Despite the many publications, the analysis of dispersion in optical systems remains complex and difficult to understand. A contributing element to the complexity is the typical focus on the polarization state and its dependence on birefringence, while the goal of dispersion analysis is often finding derivatives that determine polarization evolution. The well-known and often applied Jones matrix analysis methods for optical systems, which exclude differential analysis, are outlined in a series of papers published by Jones and Hurwitz [14]–[19]. These methods are usually applied to the analysis of polarization effects and dispersion in optical systems [21]–[25]. The power of the nondifferential Jones matrix formalism may also be a limitation. While it carries a detailed description of the polarized optical field, it renders additional complexity in dispersion calculations. It may be a factor in motivating omission of amplitude effects in many publications concerning dispersion. Unfortunately, by avoiding amplitude dispersion effects, the application of any analysis to general optical systems will lead to errors [8], [9].

As an aside, we note that in mathematical terms, the work of Jones on differential analysis [20] constituted an introduction of the Lie groups [26] to optics. The Jones matrices can be generated by using three states that are represented by the Pauli matrices

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \sigma_2 = \begin{bmatrix} 0 & -j \\ j & 0 \end{bmatrix} \qquad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The generated Jones matrices J result from a matrix exponential

$$J = \exp\left(\frac{j}{2}\mathrm{v}\cdot\sigma\right)$$

where j is an imaginary number, v is a complex vector, and σ is a triplet of Pauli matrices [27], [28]. This succinct notation is frequently used in many papers on PMD [8], [9], [11]. If the vector v is assumed to be real, the dispersion is real and the amplitude effects are excluded from the analysis. Then, the resulting Jones matrices belong to the special unitary group SU(2) [26]. If the vector v is complex, the dispersion is also complex. It will be evident in Section III that, for polarized light, the differential analysis of Jones is equivalent to the preceding formulation. The elementary matrices contain the Pauli matrices and an identity matrix that describes polarization-state independent propagation. In his paper, Jones referred to elementary matrices as Θ-matrices. The term elementary matrix was introduced later by Ramachandran *et al.* [29].

In this tutorial, we revive the theoretical framework of Jones and remain focused on his work throughout the paper as his work is conceptually straightforward. We hope that with focus on the work of Jones, we will be able to present this tutorial in an easy to understand manner. It is beyond the scope of this

Manuscript received June 01, 2009; revised August 07, 2009 and September 10, 2009. First published October 06, 2009; current version published January 27, 2010.

The authors are with the Measurement Research Laboratory, Agilent Technologies, Inc., Santa Clara, CA 95051 USA (e-mail: doug_baney@agilent.com).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/JLT.2009.2033614

tutorial to review all work on the topic of dispersion. Instead, we will focus on elementary matrix methods developed by Jones, as applied to theoretical and experimental analysis of dispersion.

Jones proposed his methods of differential analysis for longitudinal light propagation in a medium that may change its properties in a continuous manner [20]. He recognized that for certain types of problems, a method that operates on the arguments of the exponential terms that describe the optical field is more convenient than the analysis of the field itself. So, rather than working with exponential functions to describe the field, as is commonly done, the formalism focuses on the arguments of the exponential functions. While Jones was concerned with differential spatial analysis of optical systems, his methodology is directly applicable to modern-day dispersion problems that focus on frequency dependence [8], [9], [11], [30]. In this paper, we apply the differential method that Jones developed to a system with optical frequency dependence. We develop the concept of the elementary matrices and the N_ω-matrix, with eight directly measurable parameters (p's and q's) that allow straightforward calculation of dispersion, including the interaction of all eight elementary parameters that comprise group delay and amplitude slopes. We extend this approach to include the effects of concatenation of optical elements as well as to provide a simple formalism to obtain higher order PMD. We conclude with a discussion of measurement concepts for direct determination of the p and q parameters that constitute the elementary matrices and the N_ω-matrix without the need to measure the Jones matrix.

This paper is organized as follows. We discuss phase and amplitude dispersion in Section II using a scalar description of light that does not account for polarization effects, with the intent of presenting the basic concepts of the approach. In Section III, we introduce matrix notation applicable to polarized light; this leads to the optical field description based on the elementary matrices and N_ω-matrix [20]. The application of elementary matrices to the calculation of dispersion is shown in Section IV, demonstrating the importance of both phase and amplitude dispersive effects on the total effective dispersion. Section V discusses the problem of concatenation of multiple optical components, where we see that new optical properties are created as a result of the combined effects of phase and amplitude dispersion. When omitting amplitude effects, as consistent with the usual higher order PMD definition, we see in Section VI that higher order PMD can be described by simple derivatives of the elementary parameters. Measurement methods are outlined in Section VII, showing that the p and q elementary parameters are measurable with both direct detection methods and coherent interferometric arrangements. A summary follows in Section VIII, along with a discussion in the Appendix on the application of frequency-derivative elementary matrix concepts to the Mueller matrix.

II. Scalar Case of Dispersion

Consider light passing through an optical component as shown in Fig. 1. At the output terminal of the component, the light wave has an amplitude and phase that varies with frequency, as shown in Fig. 2. We refer to the change of amplitude and phase with frequency as amplitude dispersion and phase dispersion, respectively. This represents a generalization of the concept of dispersion to both phase and amplitude.

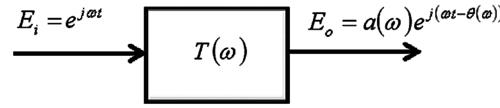

Fig. 1. Model of an optical component with transfer function $T(\omega)$ showing input and output optical fields.

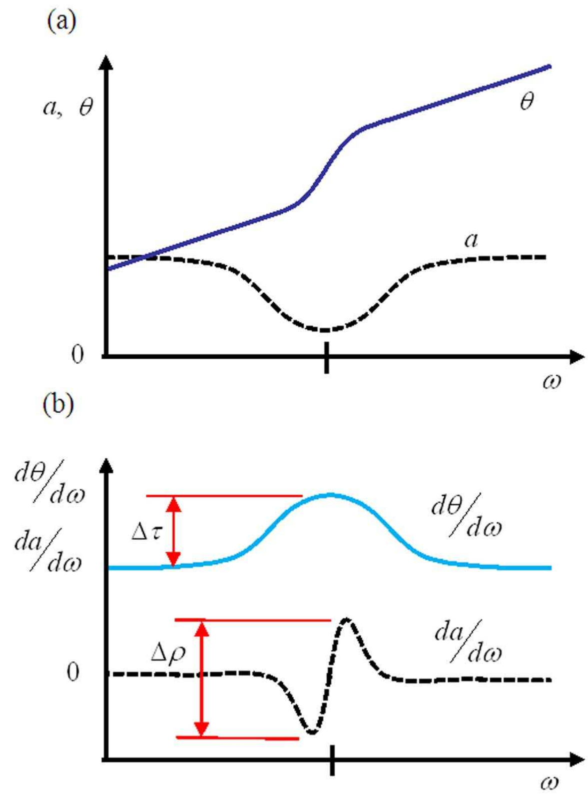

Fig. 2. (a) Illustration of the frequency-dependent phase and amplitude output of an optical component. (b) Corresponding frequency derivatives of phase and amplitudes.

The importance of an analysis that considers both phase and amplitude was recognized by Jones in his work on N-matrices applied to spatial propagation [20]. However, his work on differential operators never became popular. Amplitude effects were routinely omitted or dismissed in publications until the work of Gisin *et al.* and Huttner *et al.* appeared [8], [9]. The purpose of this section is to introduce the concept of a differential operation, similar to the function of the N_ω-matrix, for the simple case of a scalar optical wave. This provides an overview of the elementary parameter concept that characterizes the frequency sensitivity of the transfer function of a component. This approach is generalized in later sections to include the polarized optical wave and the matrix notation introduced by Jones [20]. We introduce, for the purposes of this discussion, a scalar parameter n_ω that describes the speed of change of the phase and the amplitude across a frequency interval. This provides insight into the elementary matrices and the N_ω-matrix discussed in the next section.

We describe the field at the component output as the product of three terms as

$$E_0(t,\omega) = e^{j\omega t}\, e^{-j\theta(\omega)} e^{A(\omega)} \tag{1}$$

where the first term is the time-dependent monochromatic field term with an assumed unity magnitude, the second term describes the phase change caused by the group delay of the component, and the third factor represents change in magnitude that is related to loss (or gain) of the component. Note that the field magnitude of the optical wave is represented by $a(\omega) = \exp(A(\omega))$, as shown in (1).

The effects of the component group delay and loss on the amplitude and phase of the field passing through the component are gathered into a time-invariant complex transfer function T_ω given by

$$T_\omega = e^{-j\theta(\omega)+A(\omega)}. \quad (2)$$

Then, the output optical wave can be described by the following equation:

$$E_o(\omega) = T_\omega(\omega)E_i(\omega) \quad (3)$$

where $E_i(t)$ is the monochromatic input field and T_ω is a scalar that includes both phase and amplitude effects. As we are interested in dispersion, we examine the change in the transfer function T_ω when the frequency of the incident light is shifted by $\Delta\omega$ to assess the effect on its amplitude and phase. The new transfer function at a frequency $\omega + \Delta\omega$ is given by

$$T_{\omega+\Delta\omega} = e^{-j\theta(\omega+\Delta\omega)+A(\omega+\Delta\omega)}. \quad (4)$$

The new transfer function differs from the original transfer function, T_ω, by a factor $T_{\Delta\omega}$ that describes the change in phase and amplitude as frequency changes from ω to $\omega + \Delta\omega$

$$T_{\Delta\omega} = e^{-j(\theta(\omega+\Delta\omega)-\theta(\omega))+(A(\omega+\Delta\omega)-A(\omega))}. \quad (5)$$

This equation is rewritten noting the commonly used definition of group delay $\tau = (\theta(\omega + \Delta\omega) - \theta(\omega))/\Delta\omega$. In addition, we define the amplitude slope $\rho = -(A(\omega + \Delta\omega) - A(\omega))/\Delta\omega$, so that

$$T_{\Delta\omega} = e^{-j\Delta\omega\tau-\Delta\omega\rho}. \quad (6)$$

Of interest in the preceding equation is not so much the factor $T_{\Delta\omega}$, but rather the argument of the exponential function. While $T_{\Delta\omega}$ predicts the change of the optical field, of greater interest is the argument of the exponential function that yields the dispersion-related parameters τ and ρ. Therefore, it is desirable to find a way that relates directly the change of the optical field predicted by $T_{\Delta\omega}$ to the parameters τ and ρ. The argument of the exponential function is found by taking the product of the inverse of $T_{\Delta\omega}$ and its derivative. This allows definition of a simple scalar parameter n_ω that contains τ and ρ

$$n_\omega \triangleq T'_{\Delta\omega}T^{-1}_{\Delta\omega}. \quad (7)$$

Performing the operation described by (7) on (6) yields

$$n_\omega = -j\tau - \rho. \quad (8)$$

It is clear from (8) that knowledge of n_ω leads directly to group delay and amplitude slope without any encumbrance from transcendental functions. Comparing (6) and (8), we observed that $T_{\Delta\omega}$ depends on n_ω as

$$T_{\Delta\omega} = e^{n_\omega\Delta\omega}. \quad (9)$$

So, n_ω represents the rate (scale factor) by which the phase and amplitude of the system change with frequency. When analyzing more complex problems, the ability to carry out calculations using the parameter n_ω, described by (8), rather than (6) simplifies the analysis considerably. Additionally, from (8), we immediately recognize that we can readily obtain both the phase and amplitude dispersion as the real and imaginary components of n_ω as

$$\rho = -\Re e[n_\omega] = -\frac{dA}{d\omega} \quad (10)$$

$$\tau = -\mathcal{J}m[n_\omega] = \frac{d\theta}{d\omega} \quad (11)$$

where ρ and τ are the frequency derivatives, as shown in Fig. 2 and referred to as the amplitude slope and group delay, respectively. In referring to Fig. 2(b) and (1), we note that $dA/d\omega = (1/a)(da/d\omega)$.

In the next section, we generalize this concept to the analysis of a polarized optical wave that may undergo polarization-dependent dispersion. The analysis will include the effects of retardance, absorption, birefringence, and dichroism. Following the method of Jones, we will define eight parameters $(p_0, p_1, p_2, p_3, q_0, q_1, q_2, q_3)$ that describe the amplitude and phase dispersion. The discussion of this section has already introduced two of the parameters, namely, the polarization-independent phase dispersion, denoted as p_0 with $p_0 = \tau$, and the polarization-independent amplitude dispersion q_0, with $q_0 = \rho$. Finally, rather than going through the process described by (6), we will attempt to define the differential operators, similar to n_ω, for Jones matrices. In doing so, we will follow the procedures outlined by Jones [20]. Furthermore, if we measure n_ω directly, rather than characterize the transfer function T_ω, we may further simply the dispersion analysis.

III. Polarization-Dependent Dispersion and the N_ω-Matrix

Given the vector nature of polarized light, the discussion of this section will use the vector and matrix notations developed by Jones. However, we will examine optical frequency dependence as opposed to the spatial dependence of light propagation as analyzed by Jones [20]. In his work on differential operators, Jones studied the effects of a small spatial increment Δz on the propagation of light, as shown in Fig. 3(a). Since the 2×2 Jones matrix has four complex elements, it may change in eight different ways. However, to describe the change of the optical wave, Jones selected eight physical properties of material that characterize the material slice Δz. The physical properties included retardance, absorption, birefringence, and dichroism described by eight material parameters denoted by $\eta, \kappa, \omega, \delta, \mathrm{g}_0, g_{45}, p_0$, and p_{45} [20].

These eight elementary parameters were associated with eight elementary Θ-matrices that contributed to an N-matrix describing the Jones matrix (or a Jones vector) change across a distance Δz. This elementary matrix approach was aimed at differential analysis of the optical systems, resulting in significant simplification of the analysis for cases that involved continuous change of the material spatial properties. For example, Jones provided a very simple mathematical model for the case of the twisted crystal [20].

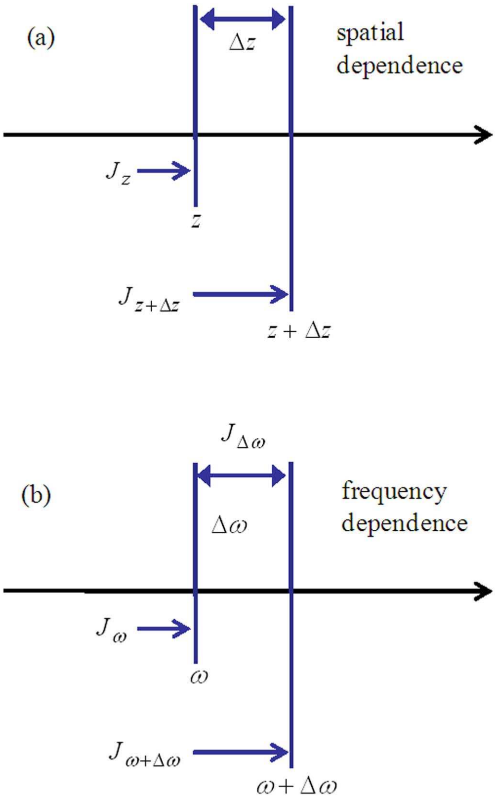

Fig. 3. (a) Jones matrices for an optical system with a spatial perturbation Δz. (b) Jones matrices corresponding to a frequency perturbation $\Delta\omega$.

Here, we apply Jones' elementary matrix methodology to the problems of phase and amplitude dispersion that have significant impact on optical communication systems. We start by examining the change in the optical field at the output of a component due to a small perturbation of optical frequency $\Delta\omega$ (instead of Δz), as shown in Fig. 3(b).

From Jones calculus, the output Jones vector E_o may be related to the input Jones vector E_i through a 2×2 Jones matrix J_ω that describes the effect of the medium on the optical field

$$E_o(\omega) = J_\omega E_i(\omega). \tag{12}$$

This is a matrix version of (3) discussed in the previous section. The optical field term $\exp(j\omega t)$ is omitted. At a frequency $\omega + \Delta\omega$, the optical field at the component output is given by

$$E_o(\omega + \Delta\omega) = J_{\omega+\Delta\omega} E_i(\omega). \tag{13}$$

From Fig. 3(b), following the usual rule of concatenation of Jones matrices, we find by inspection that

$$J_{\omega+\Delta\omega} = J_{\Delta\omega} J_\omega \tag{14}$$

where matrix $J_{\Delta\omega}$ represents the Jones matrix change due to an optical frequency change $\Delta\omega$. The matrix $J_{\Delta\omega}$ is very close to an identity matrix I, with exception for a small perturbation, due to the frequency change $\Delta\omega$, which is described by the matrix denoted as N_ω, where the subscript ω refers to the dependence on optical frequency. Thus,

$$J_{\Delta\omega} \simeq I + \Delta\omega N_\omega \tag{15}$$

with the identity matrix defined as $I \triangleq \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. By substituting (15) into (14), we obtain

$$J_{\omega+\Delta\omega} = (I + \Delta\omega N_\omega) J_\omega. \tag{16}$$

To find N_ω, we right multiply (16) by J_ω^{-1}, leading to

$$N_\omega = \frac{J_{\omega+\Delta\omega} - J_\omega}{\Delta\omega} J_\omega^{-1}. \tag{17}$$

Since the optical frequency derivative of the Jones matrix is defined as

$$J'_\omega = \lim_{\Delta\omega \to 0} \frac{J_{\omega+\Delta\omega} - J_\omega}{\Delta\omega} \tag{18}$$

(17) can be rewritten in the following form:

$$N_\omega = J'_\omega J_\omega^{-1}. \tag{19}$$

The preceding equation has a mathematical form analogous to (7). (As an aside, we note that while scalar multiplication as shown in (7) commutes, matrix operations, such as (19) do not commute, so we must always multiply matrices in the correct sequence.) The form of this equation was proposed by Jones [20]. The matrix N_ω represents the differential operator that describes the rate of change of the Jones matrix with frequency. It is a very important matrix that contains all dispersive quantities and will be used in the analysis of dispersion in the following sections. Now we examine the elements that constitute this key matrix.

Following the work of Jones, we define N_ω as the summation of eight elementary Θ-matrices that contain elementary parameters. The elementary parameters are real and define the phase and amplitude dispersion over the frequency interval $\Delta\omega$

$$N_\omega \equiv \sum_{\mathrm{i}=0}^{7} \Theta_i. \tag{20}$$

The eight elementary Θ-matrices are shown in Table I. In each matrix Θ, elementary parameters multiply a unitary matrix. The elementary parameters p_i and q_i describe the magnitude of the dispersion, while the unitary matrices determine the optical properties. The matrices that have imaginary eigenvalues describe phase change, while matrices that have real eigenvalues describe amplitude change. The matrices Θ_0 and Θ_4 describe polarization-independent media. As noted earlier, p_0 represents the group delay and q_0 represents the amplitude slope. The eigenvectors of the matrices Θ_1, Θ_2, and Θ_3 represent polarization modes that are supported in the optical media described by these matrices. Since the eigenvalues of these matrices are imaginary, the three parameters, p_1, p_2, and p_3, are components of the differential group delay (DGD). They describe the difference of the phase derivatives for three pairs of eigenvectors representing three pairs of orthogonal polarization states: (0° linear, 90° linear), (45° linear, −45° linear), and (RHC, LHC), respectively, where RHC refers to right-hand circular and LHC refers to left-hand circular polarization states. For example,

$$p_1 = \frac{1}{2}\left(\frac{d\theta_{90^\circ}}{d\omega} - \frac{d\theta_{0^\circ}}{d\omega}\right) = \frac{1}{2}(\tau_{90^\circ} - \tau_{0^\circ}). \tag{21}$$

TABLE I
ELEMENTARY PARAMETERS, DEFINITIONS, AND MEANING

Phase frequency derivatives			Amplitude frequency derivatives		
Elementary matrix	Definition	Meaning	Elementary matrix	Definition	Meaning
$\Theta_0 = -p_0 \begin{bmatrix} j & 0 \\ 0 & j \end{bmatrix}$	$p_0 = \tau = \frac{d\theta}{d\omega}$	group delay	$\Theta_4 = -q_0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$q_0 = \rho = -\frac{1}{a}\frac{da}{d\omega}$	amp. slope
$\Theta_1 = p_1 \begin{bmatrix} j & 0 \\ 0 & -j \end{bmatrix}$	$p_1 = \frac{1}{2}(\tau_{90^\circ} - \tau_{0^\circ})$	0º DGD	$\Theta_5 = q_1 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$q_1 = \frac{1}{2}(\rho_{90^\circ} - \rho_{0^\circ})$	0º DAS
$\Theta_2 = p_2 \begin{bmatrix} 0 & j \\ j & 0 \end{bmatrix}$	$p_2 = \frac{1}{2}(\tau_{-45^\circ} - \tau_{45^\circ})$	45º DGD	$\Theta_6 = q_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$q_2 = \frac{1}{2}(\rho_{-45^\circ} - \rho_{45^\circ})$	45º DAS
$\Theta_3 = p_3 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$p_3 = \frac{1}{2}(\tau_{LC} - \tau_{RC})$	circular DGD	$\Theta_7 = q_3 \begin{bmatrix} 0 & -j \\ j & 0 \end{bmatrix}$	$q_3 = \frac{1}{2}(\rho_{LC} - \rho_{RC})$	circular DAS

Similarly, as noted earlier, q_0 represents the amplitude slope. The eigenvectors of the matrices Θ_5, Θ_6, and Θ_7 are the same as the eigenvectors of Θ_1, Θ_2, and Θ_3. However, their eigenvalues are real, indicating amplitude effects. The three parameters, q_1, q_2, and $q_{3,}$ describe the difference of the amplitude derivatives for three pairs of orthogonal polarization states identified before. For example,

$$q_1 = \frac{1}{2}(\rho_{90^\circ} - \rho_{0^\circ}). \quad (22)$$

Note that, in general, the p and q parameters are frequency-dependent so that $p_i = p_i(\omega)$ and $q_k = q_k(\omega)$; however, for notation simplicity, this dependence is not shown explicitly. The reader is referred to the work of Jones to see the parallels between the analysis of the spatial dependence of light [20] and the optical frequency dependence analysis presented here. Note that p_0 and q_0 represent the case of polarization-independent propagation, while the rest of the p's and q's describe polarization-dependent propagation.

Since the polarization components (0° linear, 45° linear, and circular) may be represented on the Poincare sphere, it is convenient to think of (p_1, p_2, p_3) and (q_1, q, q_3) as vectors that represent phase and amplitude dispersion, as shown in Fig. 4. As described earlier, each vector component is proportional to the difference in optical frequency derivatives of phase or amplitude for the orthogonal polarization states defined in the Stokes space and on the Poincare sphere.

By adding up the eight matrices according to (20), we obtain the general form of the matrix N_ω as

$$N_\omega = \begin{bmatrix} q_1 - q_0 + j(p_1 - p_0) & p_3 + q_2 + j(p_2 - q_3) \\ q_2 - p_3 + j(p_2 + q_3) & -q_0 - q_1 - j(p_1 + p_0) \end{bmatrix}. \quad (23)$$

The matrix N_ω has properties that make it extremely useful for analysis of dispersion. This will be discussed in more detail in Section IV. The relationship between the matrix N_ω and the Jones matrix $J_{\Delta\omega}$ that accounts for the effects of frequency change is given by

$$J_{\Delta\omega} = e^{N_\omega \Delta\omega}. \quad (24)$$

Thus, the matrix N_ω can be used to describe the change in the Jones matrix, or the change in optical field at the component output, in response to a change in optical frequency.

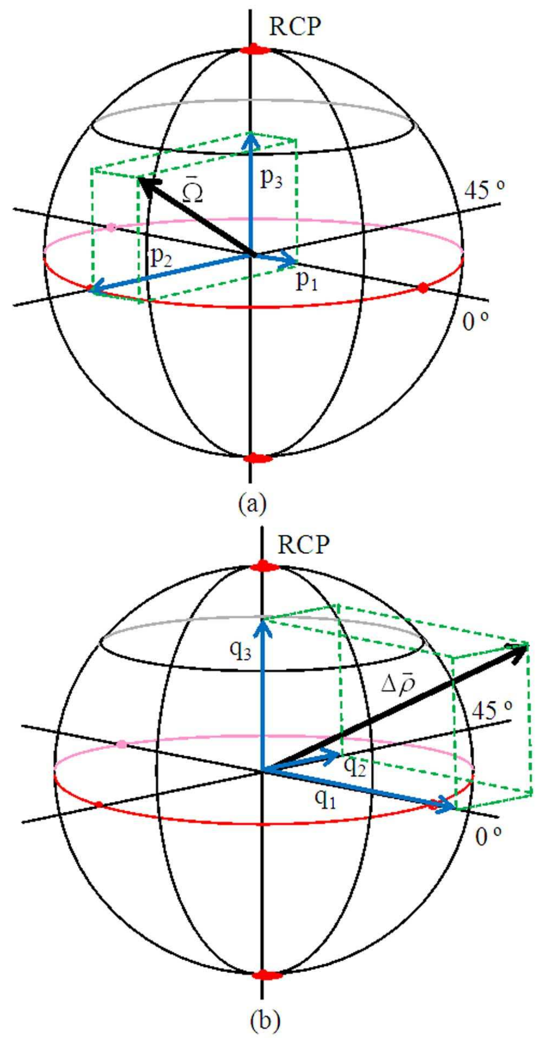

Fig. 4. Poincare representation showing (a) three components of the phase dispersion vector and (b) three components of the amplitude dispersion vector. RCP: right-circular polarization.

Example: Let us examine an exemplary elementary parameter and a Θ-matrix from Table I by considering a special case of (23). Consider the case where all elements of N_ω are zero with an exception for p_1. In this case, N_ω reduces to

$$N_\omega = \begin{bmatrix} jp_1 & 0 \\ 0 & -jp_1 \end{bmatrix}. \quad (25)$$

The Jones matrix in (24) associated with the frequency change $\Delta\omega$ is

$$J_{\Delta\omega} = \begin{bmatrix} e^{jp_1\Delta\omega} & 0 \\ 0 & e^{-jp_1\Delta\omega} \end{bmatrix}. \tag{26}$$

This is a well-known matrix that represents linear birefringence having a DGD equal to $2p_1$. The eigenvectors of N_ω and $J_{\Delta\omega}$ are the same and correspond to horizontal and vertical linear polarization states. The corresponding eigenvalues of N_ω are jp_1 and $-jp_1$. Since the eigenvalues are imaginary, the matrix N_ω changes the phase of the eigenmodes at the rates p_1 and $-p_1$ (opposite rates). Using Jones notation, the change in the field is given by

$$E(\omega+\Delta\omega) = (I + N_\omega\Delta\omega)E = J_{\Delta\omega}E. \tag{27}$$

Assuming that $E = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, (27) may also be written in the following form:

$$E(\omega+\Delta\omega) = e^{jp_1\Delta\omega}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + e^{-jp_1\Delta\omega}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{28}$$

where $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are eigenvectors (horizontal and vertical linear light). As indicated before, we see that 0° linearly polarized light undergoes an opposite phase change to that associated with the 90° linearly polarized light, confirming that $2p_1$ represents the DGD between the linear horizontal and linear vertical polarization modes. In general, this type of analysis is the easiest to express in terms of eigenvalues and eigenvectors. This will be explored in more detail in the next section. It is very important to realize that all dispersion information is contained in the matrix N_ω.

IV. Getting Dispersion From the Matrix N_ω

The definition of the N_ω-matrix, in (23), was a key result from the previous section. In this section, we will see that the matrix provides a direct path to the generalized dispersion concept that includes phase and amplitude effects for the polarized optical wave. The generalized dispersion contains a polarization-independent term and a polarization-dependent term that may be represented by a complex quantity that contains the interaction of the differential amplitude slope (DAS) with the DGD. These interactions were described in a publication by Gisin *et al.* [8]. As indicated in prior sections and illustrated with simple examples, the eigenvalues of N_ω provide information about the phase dispersion and amplitude dispersion. The complex eigenvalues indicate the changes of phase, while real eigenvalues indicate the changes of amplitude (with optical frequency ω). In Section II, we illustrated the direct connection between dispersive effects and the argument of the exponential function n_ω that can be used to describe a transfer function of an optical element as shown in (9). The notation of Section II was generalized for Jones matrices in Section III by introducing a matrix N_ω. An important property that aids in understanding why real and imaginary eigenvalues of the matrix N_ω correspond to the phase and amplitude changes of the optical wave, respectively, is the relationship between the eigenvalues of the matrix N_ω and the eigenvalues of the Jones matrix J_ω. It was pointed out by Jones [20] that the eigenvectors of the Jones matrix J_ω are the same as those of matrix N_ω, while their eigenvalues are related by

$$\Lambda_J = e^{\Lambda_N} \tag{29}$$

where Λ_N denotes the eigenvalue of the matrix N_ω and Λ_J denotes the eigenvalue of the Jones matrix J_ω. Thus, for example, imaginary eigenvalues of N_ω result in a phase change of the eigenvectors of J_ω. Furthermore, since the structure of the matrix N_ω is known, its eigenvalues predict the dispersive properties of the optical system in terms of eight elementary parameters that make this description complete. Since the structure of the matrix N_ω is always the same and always decomposable into its eight elementary contributors, it is not necessary to derive the eigenvalues of that matrix every time N_ω is computed or measured. The eight elementary parameters uniquely determine the nature of the matrix N_ω and its eigenvalues. It will be shown in Section VII that we can directly measure the elementary parameters. This not only simplifies the way we think about dispersion, but also links dispersion to its real contributors. Thus, it avoids a highly complex analysis required for systems that, for example, contain PMD and polarization-dependent loss (PDL) [8]. Nevertheless, it is required that we go through the computation of the eigenvalues of the matrix N_ω once to understand how the eight elementary parameters interact to produce the real and imaginary components of the eigenvalues. The computation of the eigenvalues Λ_N is straightforward using the standard eigenanalysis approach that starts with the computation of a determinant

$$\det(N_\omega - \Lambda_{\rm N} I) = 0. \tag{30}$$

By solving the resulting quadratic equation, we find that N_ω has two complex eigenvalues, given by

$$\begin{aligned}\Lambda_{\rm N} = &-jp_0 - q_0 \\ &\pm j\sqrt{(p_1 - jq_1)^2 + (p_2 - jq_2)^2 + (p_3 - jq_3)^2}.\end{aligned} \tag{31}$$

It is immediately obvious that the elementary parameters p_0 and q_0, which describe polarization-independent propagation of light, appear in the formula in the same way as in the scalar case described in Section II. However, the parameters that describe polarization-dependent aspects of propagation appear under the square root in a way that does not allow for separation of the DGD vector (p_1, p_2, p_3) from the DAS vector (q_1, q_2, q_3). The effective group delay (phase dispersion) and the effective amplitude slope (amplitude dispersion) are found by taking the imaginary and real parts of Λ_N, respectively. Thus, the effective group delay is

$$\begin{aligned}\tau_{\rm eff} &= {\rm Im}[\Lambda_N] \\ &= -p_0 \pm \Re e\sqrt{(p_1 - jq_1)^2 + (p_2 - jq_2)^2 + (p_3 - jq_3)^2}.\end{aligned} \tag{32}$$

We note that the p's and q's, representing phase and amplitude derivatives, interact to contribute to the effective group delay. The previous equation shows directly how the elementary parameters determine the effective group delay. In a similar manner, the effective amplitude slope is given by

$$\begin{aligned}\rho_{\text{eff}} &= \Re e[\Lambda_N] \\ &= -q_0 \pm \operatorname{Im}\sqrt{(p_1 - jq_1)^2 + (p_2 - jq_2)^2 + (p_3 - jq_3)^2}.\end{aligned} \tag{33}$$

As clearly indicated by (32) and (33), both the effective group delay and the effective amplitude slope depend on the DGD vector (p_1, p_2, p_3) and the DAS vector (q_1, q_2, q_3). Omission of these interactions may lead to errors. This may be especially important in optical systems that comprise many concatenated optical components having nonideal transmission properties. An interesting example of such systems will be provided in Section V.

To better understand the implications of (32) and (33), we consider cases of three different transmission media. In real photonic systems, the propagation of optical signals is never as simplistic as in our examples. In practice, one would expect interactions that are far more complicated; however, even the simple scenarios considered here reveal interesting and educational insights.

Case 1: There are only p's; all q's are equal to zero (no amplitude dispersion).

The effective group delay (32) simplifies since there is no DAS, and we obtain

$$\tau_{\text{eff}} = \underbrace{-p_0}_{\text{GD}} \pm \underbrace{\sqrt{(p_1)^2 + (p_2)^2 + (p_3)^2}}_{\text{PMD}}. \tag{34}$$

This is the case when the expression for group delay agrees with expressions commonly used in literature. The effective group delay includes the contributions from the polarization-independent group delay, p_0 and the DGD, $2\sqrt{p_1^2 + p_2^2 + p_3^2}$ that results from PMD. Recall that p_1, p_2, and p_3 correspond to components of a DGD that may be represented as a vector in the 3-D Stokes space, as shown in Fig. 4(a). Often, the vector (p_1, p_2, p_3) is referred to as the PMD vector, $\vec{\Omega}$. The length of this vector $2\sqrt{p_1^2 + p_2^2 + p_3^2}$ determines the magnitude of the PMD.

Case 2: All p's are zero, only q_1 is nonzero (e.g., wavelength-dependent polarizer).

Effective group delay from (32) reduces to

$$\tau_{\text{eff}} = \operatorname{Im}[\Lambda] = \pm \mathcal{R}\text{e}\sqrt{(-jq_1)^2} = 0. \tag{35}$$

Thus, as expected, the PMD is zero. We note that for PMD measurement methods that rely on the observed motion of the polarization vector on the Poincare sphere with optical frequency change, measurement error may result leading to a non-zero group delay. This is caused by mistakenly attributing all movement of the polarization state on the normalized sphere to PMD, as illustrated in Fig. 5 for the case of a wavelength-dependent polarizing element. Here the extinction ratio changes 1 dB over a frequency range of 100 GHz, yielding a q_1 value of 0.092 ps.

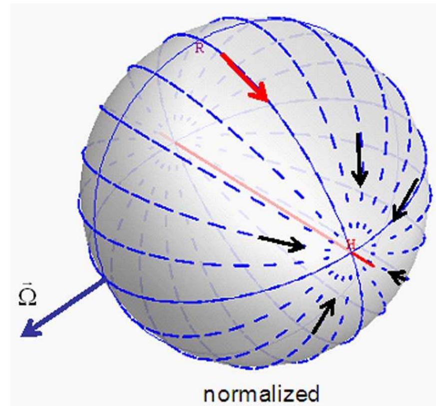

Fig. 5. Rotation of polarization state on the normalized Poincare sphere due to polarization-dependent amplitude dispersion. Misinterpretation of the polarization trajectory may lead to false identification of a real PMD vector Ω as shown. Each segment on the sphere corresponds to 100 GHz of frequency shift for a system having a 0 degree linear DAS value of $q_1 = 92$ fs.

The figure shows the change in the state of polarization over 100 GHz for a variety of input polarization states that cover the surface of the Poincare sphere. As the optical frequency changes, polarization states migrate toward the lower loss polarization state, and hence, an apparent rotation on the Poincare sphere appears even though PMD is not present.

Thus, DAS effects may induce errors in methods that rely on a measurement of polarization-state motion on the Poincare sphere with optical frequency. For example, an often-used method to measure PMD is to observe the change in polarization state at the output of a device under test as the optical frequency is scanned. The Poincare sphere methods often rely on the relationship that connects the evolution of the polarization state, represented by the Stokes vector $\vec{S}$, with the PMD vector $\vec{\Omega}$, as shown in the following equation:

$$\frac{d\,\vec{S}}{d\omega} = \vec{\Omega} \times \vec{S}\ . \tag{36}$$

We will elaborate more on this equation in the Appendix.

Case 3: All p's and q's are zero, except p_1 and q_2 with $p_1 > q_2$.

In this case, the PMD and DAS vectors are orthogonal in the Stokes space (on the Poincare sphere), and (32) and (33) for the effective group delay and the effective amplitude slope, respectively, simplify to

$$\tau_{\text{eff}} = \sqrt{p_1^2 - q_2^2} \tag{37}$$

$$\rho_{\text{eff}} = \operatorname{Im}\sqrt{p_1^2 - q_2^2} = 0. \tag{38}$$

This case is most surprising and definitely outside of the traditional understanding of dispersion. The system has a nonzero PMD and DAS; however, they interact to reduce the effective group delay, while the effective amplitude slope is equal to zero. This illustrates the advantage of the formulation presented here as DAS effects are inherently included in the dispersion relations, and the results are straightforward to analyze.

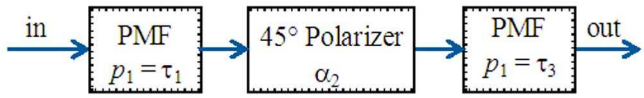

Fig. 6. Concatenated optical system composed of three elements. PMF: polarization-maintaining fiber.

V. Optical Concatenation Effects on Dispersion

Simple optical components will typically have only a few significant p and q elementary parameters out of the eight possible. We shall see that when optical components are concatenated, there is the generation of new p and q parameters giving rise to new optical effects that were not present in any of the elements comprising the concatenation. The derivation of concatenation formulas is relatively straightforward, relying on matrix and derivative properties. For simplicity, we consider a chain of only three optical elements in a concatenated sequence, as shown in Fig. 6. The generalization of the formulas to longer chains is straightforward as it relies on the same principles. We recall the relationship between the matrix N_ω and the Jones matrix first introduced in Section III

$$N_\omega = J' J^{-1}. \tag{39}$$

We know that the Jones matrix for a chain of optical components is the product of the individual Jones matrices. For three optical elements, (39) becomes

$$N_\omega = (J_3 J_2 J_1)'(J_3 J_2 J_1)^{-1}. \tag{40}$$

Applying the derivative product rule and the linear algebra rules regarding the inverse of a product of matrices, we obtain

$$N_\omega = J_3 J_2 J_1' J_1^{-1} J_2^{-1} J_3^{-1} + J_3 J_2' J_1 J_1^{-1} J_2^{-1} J_3^{-1} + J_3' J_2 J_1 J_1^{-1} J_2^{-1} J_3^{-1}. \tag{41}$$

Noting that $JJ^{-1} = I$ and rearranging terms, we get

$$N_\omega = N_{\omega,3} + J_3 N_{\omega,2} J_3^{-1} + J_3 J_2 N_{\omega,1} J_2^{-1} J_3^{-1}. \tag{42}$$

The clear pattern evident in the preceding equation suggests how to generalize the earlier formula to longer chains. The matrix $N_{\omega,3}$ that represents the last element of the chain appears in the resulting expression unchanged; however, the remaining N_ω-matrices are transformed by the Jones matrices of the optical elements that form the remainder of the chain. To illustrate some of the effects of concatenation, we consider the three element structure shown in Fig. 6. This structure contains an optical element characterized by a 0° linear birefringence described by $p_1 = \tau_1$, followed by a rotated polarizing element with a differential extinction coefficient given by α_2, and followed by a third optical element described by a 0° linear birefringence $p_1 = \tau_3$. The rotation of the second optical element transforms its original Jones matrix using the rotation matrix

$$R = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix}. \tag{43}$$

For simplicity, we chose a rotation angle, α, of 45°. The elementary matrices of birefringence are obtained from Table I, and the corresponding Jones matrices are obtained from (24). The form of the Jones matrix of the polarizing element is selected for ease of mathematical calculations. The polarizing element amplifies a horizontal linear component of light while attenuating a vertical linear component. The polarizing component is rotated by 45°. The elementary matrices and the corresponding Jones matrices for the three elements of the structure are

$$\begin{aligned} N_{\omega,1} &= \tau_1 \begin{bmatrix} j & 0 \\ 0 & -j \end{bmatrix} & J_1 &= \begin{bmatrix} e^{(j\omega\tau_1)} & 0 \\ 0 & e^{(-j\omega\tau_1)} \end{bmatrix} \\ N_{\omega,2} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} & J_2 &= R \begin{bmatrix} e^{\frac{1}{2}\alpha_2} & 0 \\ 0 & e^{\frac{-1}{2}\alpha_2} \end{bmatrix} R^{-1} \\ N_{\omega,3} &= \tau_3 \begin{bmatrix} j & 0 \\ 0 & -j \end{bmatrix} & J_3 &= \begin{bmatrix} e^{(j\omega\tau_3)} & 0 \\ 0 & e^{(-j\omega\tau_3)} \end{bmatrix}. \end{aligned} \tag{44}$$

It should be noted that since the ideal polarizing element has no wavelength dependence, its elementary matrix is a zero matrix. This illustrates an important distinction between the PDL and DAS. It is noted that since an ideal polarizer has a zero elementary matrix, it is not possible to transform it to a Jones matrix using (24). However, this does not pose a problem for the frequency-domain analysis presented in this tutorial.

By using the preceding definitions and (42), the N_ω-matrix of the chain is obtained as

$$N_\omega = \begin{bmatrix} j(\tau_3 + \tau_1 \cosh(\alpha_2)) & -je^{j2\omega\tau_3}\tau_1 \sinh(\alpha_2) \\ je^{-j2\omega\tau_3}\tau_1 \sinh(\alpha_2) & -j(\tau_3 + \tau_1 \cosh(\alpha_2)) \end{bmatrix}. \tag{45}$$

Since the aforementioned N_ω-matrix is in a different format than the N_ω-matrix of (23), it is not immediately obvious what the elementary parameter values are. However, converting the N_ω-matrix of (45) to the standard form is readily accomplished by considering the following equality:

$$\begin{bmatrix} \Delta x_{11} + j\Delta y_{11} & \Delta x_{12} + j\Delta y_{12} \\ \Delta x_{21} + j\Delta y_{21} & \Delta x_{22} + j\Delta y_{22} \end{bmatrix} = \begin{bmatrix} q_1 - q_0 + j(p_1 - p_0) & p_3 + q_2 + j(p_2 - q_3) \\ q_2 - p_3 + j(p_2 + q_3) & -q_0 - q_1 - j(p_1 + p_0) \end{bmatrix}. \tag{46}$$

By solving the previous equation, an arbitrary N_ω-matrix can be decomposed into its elementary parameters. We refer to this problem as a decomposition of the N_ω-matrix. Equation (46) is solved by gathering the corresponding real and imaginary terms of the matrices on both sides of the equation. The resulting expressions for elementary parameters are as follows:

$$\begin{aligned} p_0 &= -\frac{1}{2}(\Delta y_{11} + \Delta y_{22}) \\ p_1 &= \frac{1}{2}(\Delta y_{11} - \Delta y_{22}) \\ p_2 &= \frac{1}{2}(\Delta y_{12} + \Delta y_{21}) \\ p_3 &= -\frac{1}{2}(-\Delta x_{21} + \Delta x_{12}) \\ q_0 &= -\frac{1}{2}(\Delta x_{11} - \Delta x_{22}) \\ q_1 &= \frac{1}{2}(\Delta x_{11} + \Delta x_{22}) \\ q_2 &= \frac{1}{2}(\Delta x_{12} + \Delta x_{21}) \\ q_3 &= \frac{1}{2}(\Delta y_{21} - \Delta y_{12}). \end{aligned} \tag{47}$$

By identifying real and imaginary components of each element of the N_ω-matrix from (45) and using (47), the expressions for all elementary parameters of the chain are obtained. We note that this is a general solution that may be applied to any elementary matrix to convert it into the standard form (23). For the example at hand, the elementary parameters are given by

$$\begin{aligned} p_0 &= p_2 = p_3 = q_0 = q_1 = 0 \\ p_1 &= \tau_3 + \tau_1 \cosh(\alpha_2) \\ q_2 &= \tau_1 \ \sin(2\omega\tau_3)\sinh(\alpha_2) \\ q_3 &= \tau_1 \cos(2\omega\tau_3)\sinh(\alpha_2). \end{aligned} \tag{48}$$

We immediately note the existence of DAS terms q_2 and q_3 that were not present in any individual optical elements in the system. Their strength varies periodically with optical frequency, and is dependent on the group delays τ_1 and τ_3 and the strength of the polarizing element α_2. Clearly, the DAS terms were created due to concatenation. The values of the elementary parameters are now substituted into the equation for the effective group delay, (32), to obtain the effective group delay of the concatenated structure:

$$\tau_{\text{eff}} = \sqrt{\tau_1^2 + 2\tau_1\tau_3\cosh(\alpha_2) + \tau_3^2}. \tag{49}$$

This result brings another surprise. Noting that $\cosh(x) \equiv (e^x + e^{-x})/2$ has a minimum value of 1, the presence of PDL (or polarization-dependent gain) may result in an increase in the PMD beyond the sum $\tau_1 + \tau_3$ of the DGDs of the two birefringent elements contained in the concatenated structure. This structure was analyzed by Gisin *et al.* in a study concerned with the combined effects of PDL and PMD [8]. The results presented in this paper are the same; however, they are based on the foundation developed by Jones [20].

In Section VII, we show that it is always possible to directly measure the N_ω-matrix and the elementary parameters, and hence fully characterize dispersion. The measurement is valid even if the parameters are changing with optical frequency. However, even the precise knowledge of dispersion does not allow for time-domain analysis. The difficulties arise when the dispersion varies significantly across the modulation bandwidth of interest. If within the bandwidth occupied by the modulated optical signal the elementary parameters are approximately constant, the measurement of the N_ω-matrix can provide a good estimate of the frequency-dependent Jones matrix based on (24). Then, the response in the time domain can also be predicted. However, if the elementary parameters vary significantly with the optical frequency, then (24) is no longer applicable. This complication is known in the literature as the inverse PMD problem [13].

VI. Higher Order PMD

All elementary parameters, describing phase and amplitude effects, may depend on optical frequency. However, in considering dispersion, for many years, amplitude effects were omitted. Typically, only the phase dispersion was considered and divided into a polarization-independent part (group delay) and a polarization-dependent part (DGD). In considering DGD, closely related to PMD, it became clear that its dependence on optical frequency may produce signal impairment, particularly with high-bandwidth high-bit-rate communications [4], [13]. Thus, quantifying the frequency dependence became important and led to the definitions of higher order PMD. However, the investigation of the higher order PMD was also carried out under the assumption that the amplitude dispersion is not mixed into the total dispersion. The assumption is incorrect, as evident from (32). However, for completeness, we provide the description of the usual higher order PMD (i.e., amplitude dispersion free) in terms of elementary parameters. Thus, all equations in this section require that the DAS vector (q_1, q_2, q_3) be equal to zero. Otherwise, dispersion becomes a complex quantity that is difficult to illustrate in the Stokes space. However, in some recent publications, the general case of higher order PMD was addressed based on the use of a Magnus expansion [31].

Given the simplicity of the elementary parameters, one might ask how this extends to the calculation of higher order PMD. It turns that this calculation is straightforward. The PMD vector is often represented in Stokes space; its length corresponds to the magnitude of the PMD and its direction represents the nature of the PMD. For example, a 0° linear birefringence creates a PMD vector that only has a 0° linear component. Thus, the PMD may be represented as a vector whose components are made up of the elementary parameters, p_1, p_2,and p_3. The group delay term p_0 is independent of polarization and not included. The PMD vector is shown in Fig. 4(a) and described by the following equations:

$$\begin{aligned} \vec{\Omega} &= 2\, \vec{\tau}_{\text{eff}} = -2(p_1, p_2, p_3), \qquad \text{where} \\ |\, \vec{\tau}_{\text{eff}} \,| &= \sqrt{p_1^2 + p_2^2 + p_3^2}. \end{aligned} \tag{50}$$

The derivation of the above equation is included in the Appendix. Higher order PMD results in an optical frequency dependence of the PMD vector. Thus, existence of higher order PMD implies that one or more of the vector components vary with frequency, i.e., the derivatives with respect to frequency are nonzero. By definition, the second-order PMD is

$$\vec{\Omega}' = -2\frac{d}{d\omega}(p_1, p_2, p_3). \tag{51}$$

The *n*th-order PMD is given by

$$\frac{d^{n-1}}{d\omega^{n-1}}\, \vec{\Omega} = -2\frac{d^{n-1}}{d\omega^{n-1}}(p_1, p_2, p_3). \tag{52}$$

The ease with which first-order PMD is determined directly from the elementary matrix parameters, combined with the simple definitions of the higher order PMD, provide yet another benefit of using elementary matrices and elementary parameters. It is important to note that the preceding procedure could be easily applied to *q*-parameters describing DAS. In the next section, we shall discuss how the elementary parameters of the elementary matrices are measured.

VII. Elementary Matrix Measurements

The two most popular methods for measuring PMD are the Jones matrix eigenanalysis (JME) [21] and the phase-shift method [32], [33]. Neither of these methods, in their original form, account for all eight elementary parameters, but, instead they account for only three with JME and four with the phase-shift method. The practical difficulty with both methods is the requirement of switching the input polarization state.

Since optical components are typically characterized over a broad wavelength range, in addition to the input polarization-state switching, the optical frequency must also be swept. This introduces a complication. If the polarization states must be switched for each frequency step across the measurement range, then the measurements are very slow. If the laser is first swept across the measurement wavelength range for a constant input polarization state and then the input polarization state is switched for the next wavelength sweep, the measurement is faster; however, the precision is often degraded due to sweep-to-sweep optical frequency errors. Furthermore, we note that the JME method has difficulty measuring group delay (p_0) unless interferometric arrangements are used; however, interferometers are environmentally sensitive, thus limiting the accuracy of the group delay measurement. While the phase-shift method has the same problems, as described before, it does measure group delay directly with no phase noise corruption. This is accomplished through intensity modulation that creates two optical waves in addition to the carrier to probe the device under test (DUT). On the contrary, the elementary matrix method allows measurement of all eight dispersion parameters by using continuous polarization modulation that provides multiple optical frequencies and multiple polarization states. This allows for robust measurement against laser phase noise and frequency repeatability errors.

In this section, we will first describe, for pedagogic reasons, a conceptual approach for measurement of the elementary parameters, and then follow with a discussion of a practical measurement method. We note that there were other attempts to measure elementary parameters [30], [34]–[36]; however, so far, the idea of continuous modulation of the polarization state has not been widely explored.

From the previous sections, we have seen how the elementary matrix parameters may be used to describe the dispersive nature of a component, concatenation effects, and higher order PMD. We shall see in this section how the simple definition of the elementary matrix provides a basis for a new approach to metrology of dispersion. The significance of this new approach is that the direct measurement of the elementary parameters avoids the laborious process of measuring Jones matrices or the evolution of the polarization state on the Poincare sphere. Not only are the elementary parameter test methods often simpler, but they are also more robust as the measurements are differential in nature, and lead to rejection of phase and amplitude noise. Various methods are possible for measuring the elementary parameters based on either coherent interferometric methods or direct detection. The choice of technique may also depend on the type of component under test. For a long length of fiber, direct detection methods will be more appropriate if measurement stability becomes a significant limitation. The section contains two parts. The first part describes a conceptual approach that explores properties of the elementary matrices; the second part gives an example of a practical measurement system and explains how to use elementary matrices for the description of optical modulated signals.

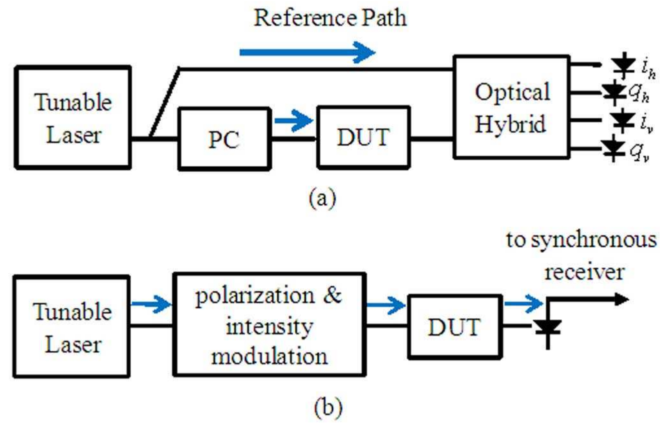

Fig. 7. Measurement setups for elementary parameter characterization with (a) interferometric arrangement and (b) optical modulation and direct-detection measurement. PC: polarization controller.

A. Measurement Principles

As mentioned in prior sections, the N_ω-matrix acts as a differential operator. By rearranging (19), the derivative of the Jones matrix can be expressed in the following form:

$$J'_\omega = N_\omega J_\omega. \tag{53}$$

Thus, by left multiplying the Jones matrix by the N_ω-matrix, the derivative of the Jones matrix can be found. This concept was introduced by Jones [20], and applied to Jones matrices and Jones vectors. For the Jones vector E, the preceding equation can be rewritten in the following form:

$$\frac{\Delta E}{\Delta \omega} \simeq N_\omega E \tag{54}$$

where the derivative of the Jones vector was replaced by a finite difference. Based on (54), it is clear that by measuring perturbations of the optical wave E for known frequency increments, it is possible to determine the matrix N_ω. Consider the interferometric arrangement shown in Fig. 7(a) containing a polarization-diverse coherent receiver. This optical receiver is capable of measuring the optical field at the output of the DUT. Suppose the polarization control is set such that the optical wave exiting the optical component under test is in the horizontal polarization state. At the optical frequency ω, the optical field is measured as $E_1(\omega)$, then the optical frequency of the tunable laser is shifted by $\Delta\omega$ and the optical wave is measured again. The new optical field is slightly different and expressed by $E_1(\omega + \Delta\omega)$. The change in the optical field at the component output is a complex vector given by

$$\Delta E_1(\Delta\omega) = E_1(\omega) - E_1(\omega + \Delta\omega) = \begin{bmatrix} \Delta a_h + j\Delta b_h \\ \Delta c_h + j\Delta d_h \end{bmatrix} \tag{55}$$

where subscript h refers to horizontal polarization state that was used in the measurement. Similarly, if the polarization controller is adjusted such that polarization state of the optical wave at the output is linear vertical, we obtain a change in the field represented by a complex vector

$$\Delta E_2(\Delta\omega) = E_2(\omega) - E_2(\omega + \Delta\omega) = \begin{bmatrix} \Delta a_v + j\Delta b_v \\ \Delta c_v + j\Delta d_v \end{bmatrix}. \tag{56}$$

Since in both cases the output optical wave was known, by using (54), the set of linear equations describing the experiment can be constructed as follows:

$$\begin{aligned} &\frac{1}{\Delta\omega} \times \begin{bmatrix} \Delta a_h + j\Delta b_h \\ \Delta c_h + j\Delta d_h \end{bmatrix} = \\ &\quad \begin{bmatrix} q_1 - q_0 + j(p_1 - p_0) & p_3 + q_2 + j(p_2 - q_3) \\ q_2 - p_3 + j(p_2 + q_3) & -q_0 - q_1 - j(p_1 + p_0) \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ &\frac{1}{\Delta\omega} \times \begin{bmatrix} \Delta a_v + j\Delta b_v \\ \Delta c_v + j\Delta d_v \end{bmatrix} = \\ &\quad \begin{bmatrix} q_1 - q_0 + j(p_1 - p_0) & p_3 + q_2 + j(p_2 - q_3) \\ q_2 - p_3 + j(p_2 + q_3) & -q_0 - q_1 - j(p_1 + p_0) \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \end{aligned} \tag{57}$$

By equating real and imaginary parts of the previous equations, all eight elementary parameters that define the N_ω-matrix can be obtained as

$$\begin{aligned} p_0 &= -(\Delta b_h + \Delta d_v)/2\Delta\omega \\ p_1 &= (\Delta b_h - \Delta d_v)/2\Delta\omega \\ p_2 &= (\Delta b_v + \Delta d_h)/2\Delta\omega \\ p_3 &= (\Delta a_v - \Delta c_h)/2\Delta\omega \\ q_0 &= -(\Delta a_h + \Delta c_v)/2\Delta\omega \\ q_1 &= (\Delta a_h - \Delta c_v)/2\Delta\omega \\ q_2 &= (\Delta a_v + \Delta c_h)/2\Delta\omega \\ q_3 &= -(\Delta b_v + \Delta d_h)/2\Delta\omega. \end{aligned} \tag{58}$$

The aforementioned procedure is just a conceptual explanation of the measurement process. In practice, the measurement is performed in a way that suppresses error sources, such as those described earlier, which exist in an experimental implementation of this conceptual experiment.

B. Practical Measurement Systems

In the aforementioned measurement illustration, we demonstrated how a simple shift in optical frequency and a measurement of the resulting perturbation of the optical wave for two polarization states may be used to determine the elementary matrix. The precision of the measurement is determined by maintaining the optical frequency shift and very accurate measurement of the perturbation of the optical wave. A very convenient way to satisfy both conditions is to use modulated optical waves. The modulation can be realized, for example, with integrated optical $LiNbO_3$ modulators that produce phase modulation, intensity modulation, or polarization modulation. The nature of the modulation is not very critical as long as the resulting sidebands have an exact frequency offset from the carrier as determined by the modulation frequency. Furthermore, the positive and the negative sidebands are typically nearly identical, i.e., they have the same amplitude, phase, and polarization states. Thus, modulation replaces the function of changing the optical frequency of the tunable laser by $\Delta\omega$, as described in the conceptual experiment in the previous section. In addition, both optical waves are available at the same time, have the same phase noise, and experience the same perturbations of phase or amplitude due to instabilities in the measurement system. Therefore, the measurement of the relative changes in the sidebands effectively provides a common-mode suppression of error sources emanating from measurement system instabilities. The most important property of the two optical waves created by the optical modulation is that their perturbations are describable in terms of elementary matrices. At the input of the DUT, the two optical sidebands are identical; however, at the output of the DUT, they both satisfy (54). For the lower frequency sideband, the change in the optical frequency is negative, while for the upper sideband, the change in the optical frequency is positive. Thus, based on (54), the optical field of the sidebands at the output of the DUT can be represented by the following equation:

$$E_{\pm i} = (I \pm i\Delta\omega N_\omega)E_0 \tag{59}$$

where i denotes the number of the sideband and E_0 represents the unperturbed optical wave at the output of the DUT. It is very important to realize the simplicity of the preceding equation. We were able to describe the two optical waves (sidebands) at the output of the DUT without following the usual laborious process of tracing the propagation of the optical waves through the DUT by using Jones matrices. The two waves were described in terms of perturbations resulting from the optical frequency shift $\Delta\omega$. The perturbation was expressed by the N_ω-matrix and all the elementary parameters contained within the N_ω-matrix. Thus, we developed a description that is directly related to dispersion and allows for a direct measurement of dispersion. Furthermore, the description is complete and contains all eight elementary parameters. However, one should remember that the optimum selection of the modulation frequency is dependent on the properties of the DUT. The modulation frequency has to be sufficiently small to maintain the validity of (59). Thus, if the DUT has sharp frequency features, a small modulation frequency must be used. On the other hand, if the DUT elementary parameters are all small and do not have strong dependence on the optical frequency, the modulation frequency can be much larger. The tradeoffs are similar in nature as with the phase-shift method [32], [33].

There are many practical implementations of the measurement systems. The measurement systems may utilize interferometric methods and heterodyne detection [37] or direction detection [38]. The experimental results in [37] clearly show that all eight elementary parameters can be separated. In particular, the group delay (phase derivative) of the absorption cell can be clearly distinguished from the amplitude derivative. The heterodyne detection of [37] offers higher dynamic range; however, if the tunable laser is swept at a high rate, the time delays in long DUTs, such as fiber spools, may result in high-frequency heterodyne beat tones that increase the required receiver bandwidth. For this reason, direct detection methods are more suitable for long DUTs [38]. The detailed description of these measurement systems is beyond the scope of this paper; however, for completeness, we provide an overview of the direct detection method described in [38]. For more details, the reader is referred to prior publications on this subject [37], [38].

In a prior discussion, it was explained how the creation of sidebands through optical modulation is equivalent to shifting the optical frequency by $\pm\Delta\omega$. The remaining topics that need to be addressed are: how to control polarization state and how to measure the small perturbations of the optical wave. The easiest approach to polarization-state control is to generate multiple sideband pairs in different polarization states at modulation frequency $\Delta\omega$ [37]. Then, two pairs of sidebands can be measured at the same time, providing all the required information to solve for all elementary parameters. An alternative way is to sweep polarization state along a trajectory on the Poincare sphere at a relatively low rate [38]. The trajectory is selected to have a center of mass at the center of the Poincare sphere, i.e., it weighs all polarization states equally. This type of implementation is similar to the traditional phase-shift method, with the exception that instead of switching polarization states, the polarization state is continuously swept. The simplified diagram of this system is shown in Fig. 7(b). Light from a tunable laser undergoes intensity modulation and polarization sweeping that covers a trajectory of choice. The intensity modulation produces sidebands at $\pm\Delta\omega$. The modulation frequency may range from hundreds of megahertz to few gigahertz. Polarization modulation is at a lower frequency (few kilohertz). The optical sidebands at $\pm\Delta\omega$ propagate through the DUT and experience opposite perturbations, as described by (59). The optical signal from the DUT is detected and demodulated synchronously at the frequency $\Delta\omega$. The synchronous demodulation provides a complex signal having amplitude and phase. As shown in [38], the phase of the demodulated signal can be expressed in terms of elementary parameters. Thus, given the multiplicity of polarization states created by the polarization sweeping, the elementary parameters can be obtained. Fig. 8 shows an exemplary measurement taken with this technique on a 50 km spool of SMF-28 single-mode fiber. The results show an extremely low noise level of 10 fs as a result of the measurement system immunity from environmentally induced fluctuations of the optical waves. The stability of the system allows monitoring the evolution of the PMD over time. As illustrated in Fig. 8 in a laboratory environment, the DGD minimum shifted over 15 nm in 5 h.

VIII. Summary

Elementary matrix methods offer a powerful set of tools for characterizing optical components and systems. With the eight elementary parameters that constitute the N-matrix, there is a complete description of the phase and amplitude effects and the interaction between them. While affording less mathematical complexity, elementary matrices and parameters offer additional insights into both phase and amplitude dispersion, as required for complete characterization of the optical dispersion. Adaptation of the elementary matrix method, proposed by Jones for analysis of spatial propagation of light, to the analysis of frequency-dependent propagation allows for complete analysis and measurement of dispersion including all polarization effects.

In this paper, we defined the matrix N_ω with eight constituent elementary parameters, and derived the general relationships for the effective group delay and the effective amplitude slope. This allowed straightforward calculation of dispersion that includes the effects of amplitude slope (amplitude dispersion) and group delay (phase dispersion). The simplicity of the approach results from the differential nature of the N_ω-matrix that captures the argument of the exponential functions that are typically used for description of the optical field. This avoids complexity that arises when dealing with a full Jones matrix notation that can mask insight and lead to simplifications and omission of important optical effects.

We have also shown how to determine the elementary matrix of a concatenation of optical components in an optical system. The effects of concatenation may generate new elementary parameters that are not present in any of the individual constitutive components. This is especially important in systems that contain both PDL and PMD. We demonstrated that the conventional real higher order PMD can be described by the frequency derivatives of $p_1(\omega), p_2(\omega),$ and $p_3(\omega)$, providing a simple and elegant way of describing the frequency dependence of the PMD vector.

Additionally, we provided an overview of conceptual and practical measurement techniques that allow a direct determination of the elementary parameters. A brief comparison with the JME and phase-shift method was included. The practical implementations we described were based on modulated optical systems that contained multiple optical sidebands. We have shown that the perturbations of the optical sidebands are directly proportional to the N_ω -matrices.

While the focus of this paper was on the elementary matrix methods introduced by Jones, in the Appendix, we show how similar concepts may also be applied to Mueller matrices. This links the elementary parameters and matrices with methods that rely on the evolution of the polarization state on the Poincare sphere. The analysis shows how the Poincare sphere methods can be generalized to include amplitude dispersion in addition to phase dispersion.

Appendix
Mueller Elementary Matrix

Mueller matrix methods are often used in the analysis and measurement of optical systems. The strength of the Mueller methods is the ease of visualization of optical phenomena in the Stokes space, often, on the Poincare sphere. The polarized optical waves are described by Stokes vectors. For monochromatic light, there is a unique mapping between the Jones space and the Stokes space [28], [39]. In 1978, inspired by the works of Jones [20], Azzam proposed a differential matrix calculus using elementary matrices appropriate for use with Mueller matrices [40]. Just like Jones, Azzam applied his analysis to the spatial propagation of light. Here, we adapt the differential description of the spatial propagation, developed by Azzam [40], to a differential analysis of the optical frequency dependence of Mueller matrices. The analysis introduces the m-matrix that acts as a differential operator on a Mueller matrix [34], [41]. Thus, it provides a very convenient description of the dispersive effects in Stokes space. We also note that it is possible to express the

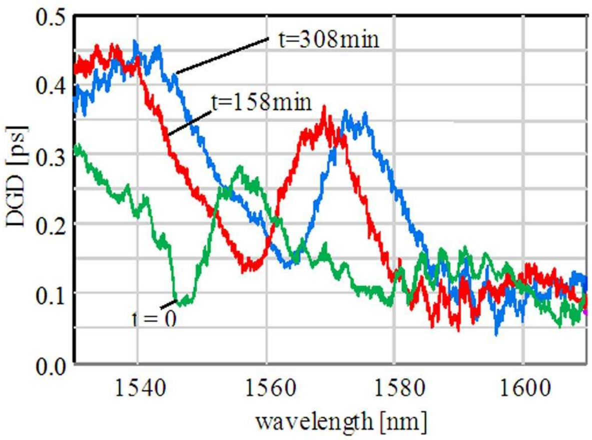

Fig. 8. DGD measurement of a 50 km spool of SMF-28 single-mode optical fiber.

m-matrix using the same elementary parameters, p and q, which were used in the discussion of the Jones N-matrix [28]–[41].

Following the logic of (54), we define the frequency derivative of the Stokes vector as the product of the Mueller m-matrix m_ω and the Stokes vector $\vec{S}$ as

$$\frac{d\vec{S}}{d\omega} = m_\omega \vec{S} \tag{A1}$$

where $\vec{S}$ is the usual Stokes vector (s_0, s_1, s_2, s_3) composed of four elements describing the total power and powers contained in polarization states defining the Stokes space, i.e., 0° linear, 90° linear, and circular polarization states. The Mueller m-matrix may be written in terms of the elementary parameters, p and q, as follows:

$$m_\omega = 2\begin{bmatrix} -q_0 & q_1 & q_2 & q_3 \\ q_1 & -q_0 & p_3 & -p_2 \\ q_2 & -p_3 & -q_0 & p_1 \\ q_3 & p_2 & -p_1 & -q_0 \end{bmatrix}. \tag{A2}$$

We note, in comparing m_ω to N_ω [(23)], that all the elements of the m_ω are real. Additionally, we note the absence of p_0. The parameter p_0, which represents group delay, is not a part of the m-matrix because all elements of the Stokes vector represent power. The elementary m-matrix concept was used recently to directly measure elementary parameters in a method that was called a generalized Muller method [34], [35].

We now use this complete m-matrix to derive a commonly used relationship that describe the evolution of the polarization state on the Poincare sphere [(36)] due to the PMD of a component (or fiber) [1]. If we assume that the amplitude slope, q_0, and the DASs, q_1, q_2, q_3, are all zero, then m_ω is reduced to a 3×3 matrix without loss of any information:

$$m_\omega = 2\begin{bmatrix} 0 & p_3 & -p_2 \\ -p_3 & 0 & p_1 \\ p_2 & -p_1 & 0 \end{bmatrix}. \tag{A3}$$

Therefore, (A1) can be expressed in the following form:

$$\frac{d\vec{S}}{d\omega} = 2\begin{bmatrix} 0 & p_3 & -p_2 \\ -p_3 & 0 & p_1 \\ p_2 & -p_1 & 0 \end{bmatrix}\vec{S}. \tag{A4}$$

Noting that the cross product between the vector $\vec{a} = (a_1, a_2, a_3)$ and the vector $\vec{b} = (b_1, b_2, b_3)$ can be expressed as the matrix multiplication as follows:

$$\vec{a} \times \vec{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}\vec{b}. \tag{A5}$$

We immediately see that (A4) can be rewritten as

$$\frac{d\vec{S}}{d\omega} = \vec{\Omega} \times \vec{S} \tag{A6}$$

where $\vec{S} = (s_1, s_2, s_3)$ is the Stokes vector that contains only polarization-state-related components and $\vec{\Omega} = -2(p_1, p_2, p_3)$ is our usual PMD vector. As discussed in Section VI, the relationship given in (A6) is often used to determine PMD by observing polarization-state evolution on the Poincare sphere. Clearly, based on discussion in this Appendix, the method based on (A6) is only valid under the assumption that all elementary parameters q are equal to zero. This assumption is in error in systems where amplitude dispersion (i.e., DAS) is present. It is also clear that the method can be easily generalized to include amplitude dispersion by using the m-matrix of (A1) and (A2).

References

[1] C. D. Poole and T. E. Darcie, "Distortion related to polarization-mode dispersion in analog lightwave systems," *J. Lightw. Technol.*, vol. 11, no. 11, pp. 1749–1759, Nov. 1993.

[2] N. Gisin and J. P. Pellaux, "Polarization mode dispersion: Time versus frequency domain," *Opt. Commun.*, vol. 89, pp. 316–323, 1992.

[3] E. Iannone, F. Matera, A. Galtarossa, G. Gianello, and M. Schiano, "Effect of polarization dispersion on the performance in IM-DD communication systems," *Photon. Technol. Lett.*, vol. 5, no. 10, pp. 1247–1249, 1993.

[4] H. Kogelnik, L. E. Nelson, and R. M. Jopson, "Polarization mode dispersion impairment in lightwave transmission systems," in *Proc. Opt. Fiber Commun. Conf. (OFC 2002)*, p. 194, Tutorial WD.

[5] P. K. Kondamuri, C. Allen, and D. L. Richards, "Variation of PMD-induced outage rates and durations with link length on buried standard single-mode fibers," in *Proc. Opt. Fiber Commun. Conf.*, 2005, pp. 1–3, Paper OThX3.

[6] F. Bruyère, "Impact of first- and second-order PMD in optical digital transmission systems," *Opt. Fiber Technol.*, vol. 2, pp. 269–280, 1996.

[7] D. Wang and C. R. Menyuk, "Calculation of penalties due to polarization effects in a long-haul WDM system using a stokes parameter model," *J. Lightw. Technol.*, vol. 19, no. 4, pp. 487–494, Apr. 2001.

[8] N. Gisin and B. Huttner, "Combined effects of polarization mode dispersion and polarization dependent losses in optical fibers," *Opt. Commun.*, vol. 142, pp. 119–125, 1997.

[9] B. Huttner, C. Geiser, and N. Gisin, "Polarization-induced distortions in optical networks with polarization-mode dispersion and polarization-dependent losses," *IEEE J. Sel. Topics Quantum Electron.*, vol. 6, no. 2, pp. 317–329, Mar./Apr. 2000.

[10] V. Chernyak, M. Chertkov, I. Gabitov, I. Kolokolov, and V. Lebedev, "PMD-induced fluctuations of bit-error rate in optical fiber systems," *J. Lightw. Technol.*, vol. 22, no. 4, pp. 1155–1168, Apr. 2004.

[11] J. P. Gordon and H. Kogelnik, "PMD fundamentals: Polarization mode dispersion in optical fibers," in *Proc. Nat. Acad. Sci. USA*, Apr. 2000, vol. 97, pp. 4541–4550.

[12] P. Williams, "PMD measurement techniques and how to avoid the pitfalls," *J. Opt. Fiber Commun.*, vol. 1, pp. 84–105, 2004, Rep. 1.

[13] H. Kogelnik, L. E. Nelson, and J. P. Gordon, "The inverse PMD problem," *J. Opt. Fiber Commun.*, vol. 1, pp. 107–122, 2004, Rep. 1.

[14] R. C. Jones, "A new calculus for the treatment of optical systems I. Description and discussion of the calculus," *J. Opt. Soc. Amer.*, vol. 31, pp. 488–493, Jul. 1941.

[15] H. Hurwitz Jr. and R. C. Jones, "A new calculus for the treatment of optical systems II. Proof of the three general equivalence theorems," *J. Opt. Soc. Amer.*, vol. 31, pp. 493–499, Jul. 1941.

[16] R. C. Jones, "A new calculus for the treatment of optical systems III. The sohncke theory of optical activity," *J. Opt. Soc. Amer.*, vol. 31, pp. 500–503, Jul. 1941.

[17] R. C. Jones, "A new calculus for the treatment of optical systems, IV," *J. Opt. Soc. Amer.*, vol. 32, pp. 486–493, Aug. 1942.

[18] R. C. Jones, "A new calculus for the treatment of optical systems, V. A more general formulation and description of another calculus," *J. Opt. Soc. Amer.*, vol. 37, no. 3, pp. 107–110, Feb. 1947.

[19] R. C. Jones, "A new calculus for the treatment of optical systems, VI. Experimental determination or the matrix," *J. Opt. Soc. Amer.*, vol. 37, no. 2, pp. 110–112, Feb. 1947.

[20] R. C. Jones, "A new calculus for the treatment of optical systems. VII properties of the N-matrices," *J. Opt. Soc. Amer.*, vol. 38, no. 8, pp. 671–685, Aug. 1948.

[21] B. L. Heffner, "Automated measurement of polarization mode dispersion using Jones matrix eigenanalysis," *IEEE Photon. Technol. Lett.*, vol. 4, no. 9, pp. 1066–1069, Sep. 1992.

[22] A. E. Willner, S. M. R. M. Nezam, L. Yan, Z. Pan, and M. C. Hauer, "Monitoring and control of polarization-related impairments in optical fiber systems," *J. Lightw. Technol.*, vol. 22, no. 1, pp. 106–125, Jan. 2004.

[23] D. Penninckx and V. Morénas, "Jones matrix of polarization mode dispersion," *Opt. Lett.*, vol. 24, no. 13, pp. 875–877, 1999.

[24] E. Forestieri and L. Vincetti, "Exact evaluation of the jones matrix of a fiber in the presence of polarization mode dispersion of any order," *J. Lightw. Technol.*, vol. 19, no. 12, pp. 1898–1909, Dec. 2001.

[25] H. Kogelnik, L. E. Nelson, J. P. Gordon, and R. M. Jopson, "Jones matrix for second order polarization mode dispersion," *Opt. Lett.*, vol. 25, pp. 19–21, 2000.

[26] G. Arfken, *Mathematical Methods for Physicists*. San Diego, CA: Academic, 1985.

[27] C. Whitney, "Pauli-algebraic operators in polarization optics," *J. Opt. Soc. Amer.*, vol. 61, pp. 1207–1213, 1971.

[28] C. Brosseau, *Fundamentals of Polarized Light: A Statistical Optics Approach*. New York: Wiley, 1998.

[29] G. N. Ramachandran and S. Ramaseshan, , S. Flügge, Ed., *Crystal Optics in Handbuch der Physik*. Berlin, Germany: Springer-Verlag, 1961, vol. XXV/I.

[30] T. Do, H. Dong, and P. Shum, "Measurement of the complex polarization mode dispersion vector using Jones matrix analysis," *Appl. Opt.*, vol. 46, pp. 916–921, 2007.

[31] M. Reimer, D. Yevick, and D. Dumas, "The accuracy of the magnus expansion for polarization-mode dispersion and polarization-dependent loss," *J. Lightw. Technol.*, vol. 16, no. 19, pp. 3337–3344, Oct. 2008.

[32] P. A. Williams and J. D. Kofler, "Narrow-band measurement of differential group delay by a six-state RF phase-shift technique: 40 fs single-measurement uncertainty," *J. Lightw. Technol.*, vol. 22, no. 2, pp. 448–456, Feb. 2004.

[33] P. A. Williams, "Modulation phase-shift measurements of PMD using only four launched polarization states: A new algorithm," *Electron. Lett.*, vol. 34, pp. 1578–1579, Sep. 1999.

[34] H. Dong, P. Shum, M. Yan, J. Q. Zhou, G. X. Ning, Y. D. Gong, and C. Q. Wu, "Generalized Mueller matrix method for polarization mode dispersion measurement in a system with polarization-dependent loss or gain," *Opt. Exp.*, vol. 14, pp. 5067–5072, 2006.

[35] H. Dong, P. Shum, Y. D. Gong, M. Yan, J. Q. Zhou, and C. Q. Wu, "Virtual generalized mueller matrix method for measurement of complex polarization-mode dispersion vector in optical fibers," *Photon. Technol. Lett.*, vol. 19, no. 1, pp. 27–29, Jan. , 2007.

[36] L. Chen, O. Chen, S. Hadjifaradji, and X. Bao, "Polarization-mode dispersion measurement in a system with polarization-dependent loss or gain," *Photon. Technol. Lett.*, vol. 16, pp. 206–208, 2004.

[37] B. Szafraniec and D. M. Baney, "Elementary matrix-based vector optical network analysis," *J. Lightw. Technol.*, vol. 25, no. 4, pp. 1061–1069, Apr. 2007.

[38] B. Szafraniec, R. Mästle, and D. M. Baney, "Polarization mode dispersion measurement based on continuous polarization modulation," *Appl. Opt.*, vol. 47, no. 7, pp. 1–9, 2008.

[39] J. W. Hovenier, "Structure of a general pure Mueller matrix," *Appl. Opt.*, vol. 33, pp. 8318–8324, 1994.

[40] R. M. A. Azzam, "Propagation of partially polarized light through anisotropic media with or without depolarization: A differential 44 matrix calculus," *J. Opt. Soc. Amer. B, Opt. Phys.*, vol. 68, no. 12, pp. 1756–1767, 1978.

[41] D. M. Baney, "Complex component characterization based on elementary matrices," in *Proc. Optical Fiber Communications Conf. (OFC 2009)*, San Diego, CA, 2009, Tutorial OThB.

Douglas M. Baney received the B.S.E.E. degree from California Polytechnic State University, San Luis Obispo, in 1981, the M.S.E.E. degree from the University of California, Santa Barbara, in 1986, and the Ph.D. degree from the Ecole Nationale Superieure des Telecommunications, Paris, France, in 1990.

In 1981, he joined Hewlett-Packard as an R&D Engineer, where he designed hybrid GaAs microwave circuits and singlemode fiber-optic instrumentation. In 1991, he was transferred to HP's Laboratories, where he developed new techniques for optical amplifier noise characterization, visible upconversion fiber lasers, and reflectometry, and later managed projects on coherent optical telecommunications instrumentation and laser-based optical mice for computers. He is currently a Department Manager in the Measurement Research Laboratory, Agilent Technologies, Santa Clara, CA, where he is responsible for applied research in microwave and optical measurement, distributed sensing, and software systems.

Dr. Baney is the Optical Society of America representative on the Optical Fiber Communication Conference and Exposition (OFC)/National Fiber Optic Engineers Conference (NFOEC) Steering Committee. He was the General Co-Chair of the Optical Fiber Communications Conference, the General Co-Chair of the Optical Amplifiers and Their Applications Conference, and a Guest Editor for the *Journal of Lightwave Technology*. He was a member on the Tyndall Award Selection Committee.

Bogdan Szafraniec was born in Sokolow Podlaski, Poland, in 1956. He received the M.S. and Ph.D. degrees in electrical engineering from Illinois Institute of Technology, Chicago, in 1982 and 1988, respectively.

He was an Electronics Designer of crystal and *LC* filters, matrix video switches, and analog fiber-optic links. In 1989, he joined Honeywell Technology Center, Phoenix, AZ, where he was involved with fiber-optic gyroscopes and their components, including integrated optics, phase modulators, sensing coils, and fiber light sources, and played a major role in the development of depolarized gyros. In 1999, he joined Agilent Technologies, Santa Clara, CA, where he investigates technologies for new optical test instruments, and is currently with the Measurement Research Laboratory. He was involved with coherent optical spectrum and network analyzers. In network analysis, he explored measurement techniques immune to environmental effects. His research interests include complex optical modulation analysis.

Dr. Szafraniec is a member of the Optical Society of America.